

TOPIC

SUMMARY LESSON

UNIT

1

1. Kinetic Theory of Ideal Gas
2. First Law of Thermodynamic
3. Engine

UNIT

2

1. Superposition Principle and Standing Wave
2. Interference and Diffraction

UNIT

3

1. Magnetic Field
2. Induced Voltages and Inductance
3. Self-Induction
4. Alternating Current



- 1 **Kinetic Theory of Ideal Gases**
- 2 **Temperature and the Zeroth Law of Thermodynamic**
- 3 **Thermometers and Temperature Scales**
- 4 **Macroscopic Description of an Ideal Gas**
- 5 **The Kinetic Theory of Gases**
- 6 **Summary**

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Lesson1 Kinetic Theory of Ideal Gases



Kinetic Theory of Ideal Gases

Pressure in Kinetic Theory of Ideal Gases

$$P = \frac{2}{3} \left(\frac{N}{V} \right) \left(\frac{1}{2} m_0 (v^2)_{av} \right)$$

Pressure of an ideal gas exerted on the wall

$$P = \frac{F}{A}, F = \frac{\Delta P}{\Delta t}, \Delta p = N m_0 \Delta v$$

Elastic collision

$$P = \frac{F}{A} = \frac{\Delta p}{A \Delta t} = \frac{2 N m_0 v}{A \Delta t}$$

Inelastic collision:

$$P = \frac{F}{A} = \frac{\Delta p}{A \Delta t} = \frac{N m_0 v}{A \Delta t}$$

Ideal Gases equation

$$PV = nRT, PV = N k_B T$$

Combined gas law

$$\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2}$$

Isobaric process (P = Const)

$$\frac{V_1}{T_1} = \frac{V_2}{T_2}$$

Isochoric process (V = Const)

$$\frac{P_1}{T_1} = \frac{P_2}{T_2}$$

Isothermal process (T = Const)

$$P_1 V_1 = P_2 V_2$$

Average kinetic Energy of each molecules.

$$K_{av} = \frac{1}{2} m_0 (v^2)_{av} = \frac{3}{2} \frac{PV}{N} = \frac{3}{2} k_B T$$

Total average kinetic Energy of molecules.

$$K = N K_{av} = \frac{3}{2} N k_B T = \frac{3}{2} n R T = \frac{3}{2} P V$$

Root-mean-square speed or v_{rms} speed

$$v_{rms} = \sqrt{(v^2)_{av}} = \sqrt{\frac{3 k_B T}{m_0}} = \sqrt{\frac{3 R T}{M}} = \sqrt{\frac{3 P}{\rho}}$$

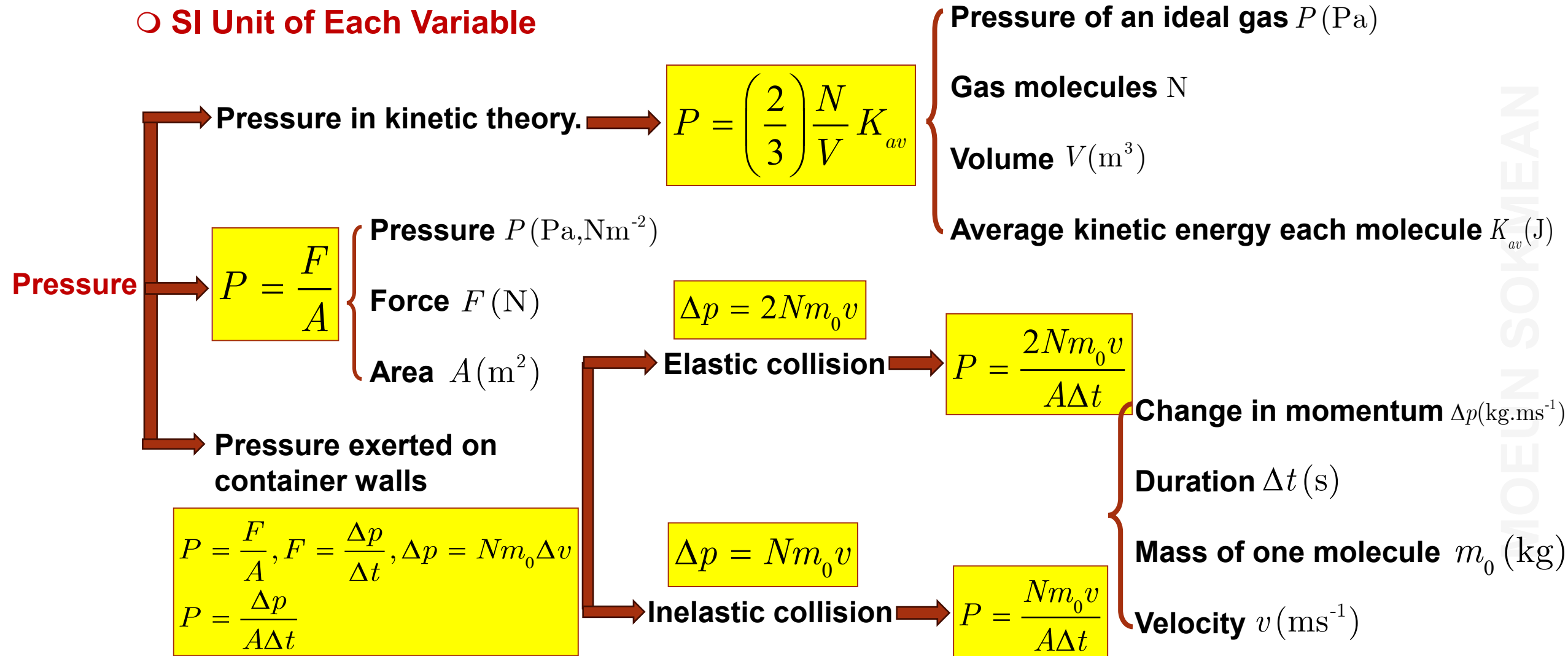
$$\overline{v} \neq \overline{v^2} \therefore \overline{v} = \frac{v_1 + v_2 + v_3 + \dots + v_n}{N}$$



Lesson1 Kinetic Theory of Ideal Gases



○ SI Unit of Each Variable



Lesson1 Kinetic Theory of Ideal Gases



○ SI Unit of Each Variable

Ideal Gas Equation

$$PV = nRT$$

$$PV = Nk_B T$$

Combined gas law
equation

$$\frac{PV_1}{T_1} = \frac{PV_2}{T_2} = cte$$

Isobaric Process $P = cte$

$$\frac{V_1}{T_1} = \frac{V_2}{T_2} = cte$$

Isochoric Process $V = cte$

$$\frac{P_1}{T_1} = \frac{P_2}{T_2} = cte$$

Isothermal Process $T = cte$

$$P_1V_1 = P_2V_2 = cte$$

Gas pressure $P(\text{Nm}^{-2}, \text{Pa})$

Number of molecules N

Number moles $n(\text{mol})$

Volume $V(\text{m}^3)$

Absolute temperature $T(\text{K})$

Boltzmann constant $k_B = 1.38 \times 10^{-23} \text{J/K}$

Universal Gas Constant

$$R = k_B N_A = 8.31 \text{J/mol.K}$$

$$P_1; P_2(\text{Pa}, \text{atm}....)$$

$$V_1; V_2(\text{m}^3, \text{dm}^3, \text{l}....)$$

$$T_1; T_2(\text{K})$$

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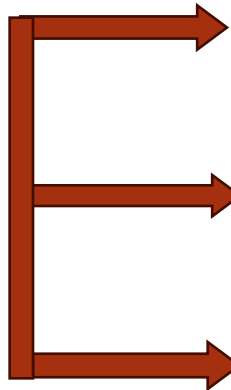


Lesson1 Kinetic Theory of Ideal Gases



○ SI Unit of Each Variable

Average kinetic energy of each molecule

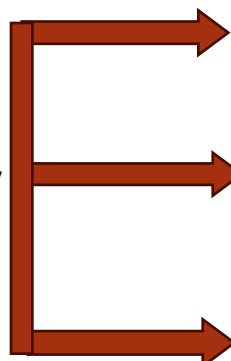


$$K_{av} = \frac{1}{2} m_0 \overline{v^2}$$

$$K_{av} = \frac{3}{2} \frac{PV}{N}$$

$$K_{av} = \frac{3}{2} k_B T$$

Total kinetic energy



$$K = NK_{av}$$

$$K = \frac{3}{2} Nk_B T$$

$$K_{av} = \frac{3}{2} nRT = \frac{3}{2} PV$$

Root-Mean Square Speed



$$v_{rms} = \sqrt{\overline{v^2}} = \sqrt{\frac{3k_B T}{m_0}} = \sqrt{\frac{3RT}{M}}$$

Average kinetic energy K_{av} (J)

Total kinetic energy K (J)

Gas pressure P (Nm⁻², Pa)

Number of molecules N

Number of moles n (mol)

Volume V (m³)

Mass of one molecule m_0 (kg)

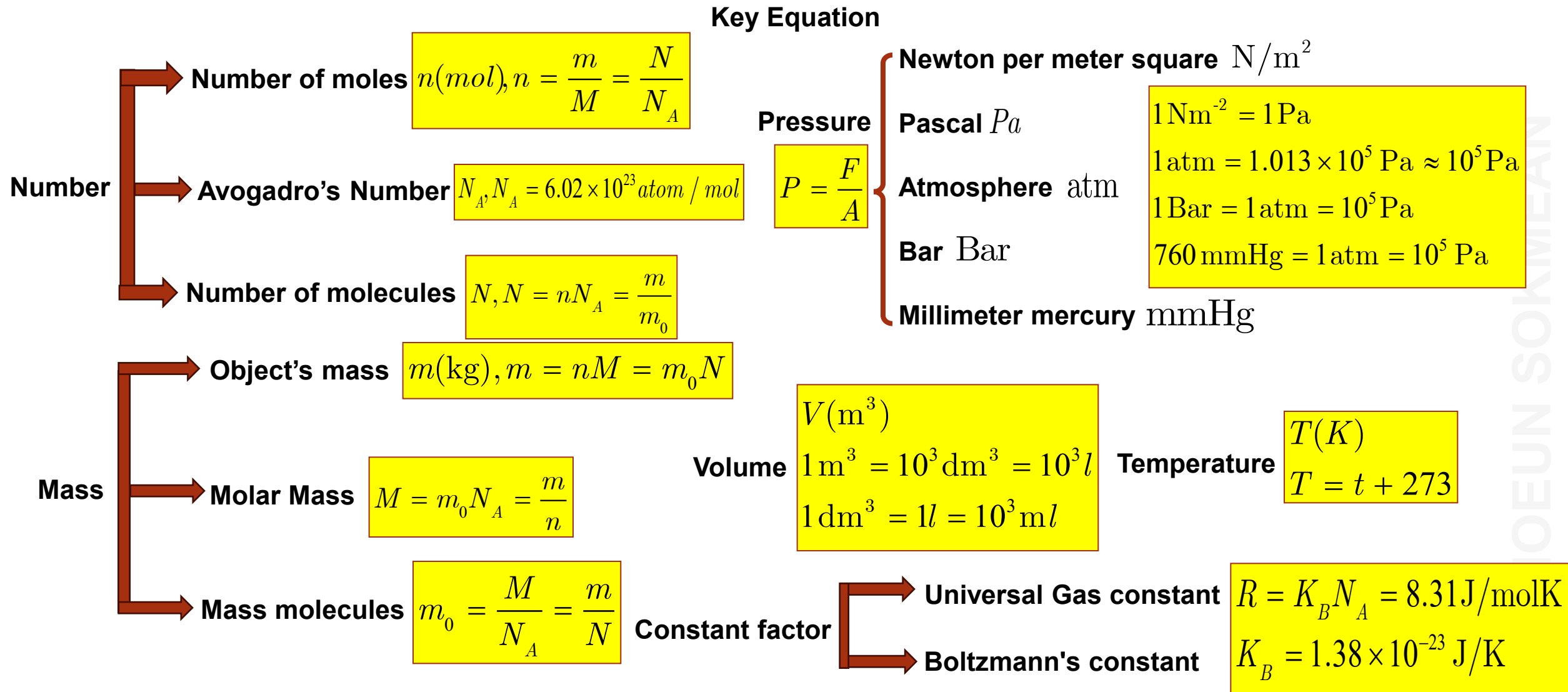
Molar Mass M (kg)

Average speed v (m/s)

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Lesson1 Kinetic Theory of Ideal Gases



Lesson1 Kinetic Theory of Ideal Gases



Fundamental Relation



Number



Number of moles

$$n(\text{mol}), n = \frac{m}{M} = \frac{N}{N_A}$$



Number of molecules

$$N, N = nN_A = \frac{m}{m_0}$$



Avogadro's numbers

$$N_A, N_A = 6.02 \times 10^{23} / \text{mol}$$

Mass



Molar Mass

$$M = m_0 N_A = \frac{m}{n}$$



Mass molecules

$$m_0 = \frac{M}{N_A} = \frac{m}{N}$$



Object's mass

$$m(\text{kg}), m = nM = m_0 N$$

Pressure N/m^2



$$P = \frac{F}{A} \left\{ \begin{array}{l} \text{Force } F(\text{N}) \\ \text{Area } A(\text{m}^2) \end{array} \right.$$

$$1 \text{ Nm}^{-2} = 1 \text{ Pa}$$

$$1 \text{ atm} = 1.013 \times 10^5 \text{ Pa} \approx 10^5 \text{ Pa}$$

$$1 \text{ Bar} = 1 \text{ atm} = 10^5 \text{ Pa}$$

Volume (m^3)



$$1 \text{ m}^3 = 10^3 \text{ dm}^3 = 10^3 \text{ l}$$

$$1 \text{ dm}^3 = 1 \text{ l} = 10^3 \text{ ml}$$

Constant coefficient



Universal's constant

$$R = K_B N_A = 8.31 \text{ J/molK}$$



Boltzmann's constant

$$k_B = 1.38 \times 10^{-23} \text{ J/K}$$

Temperature $T(\text{K})$

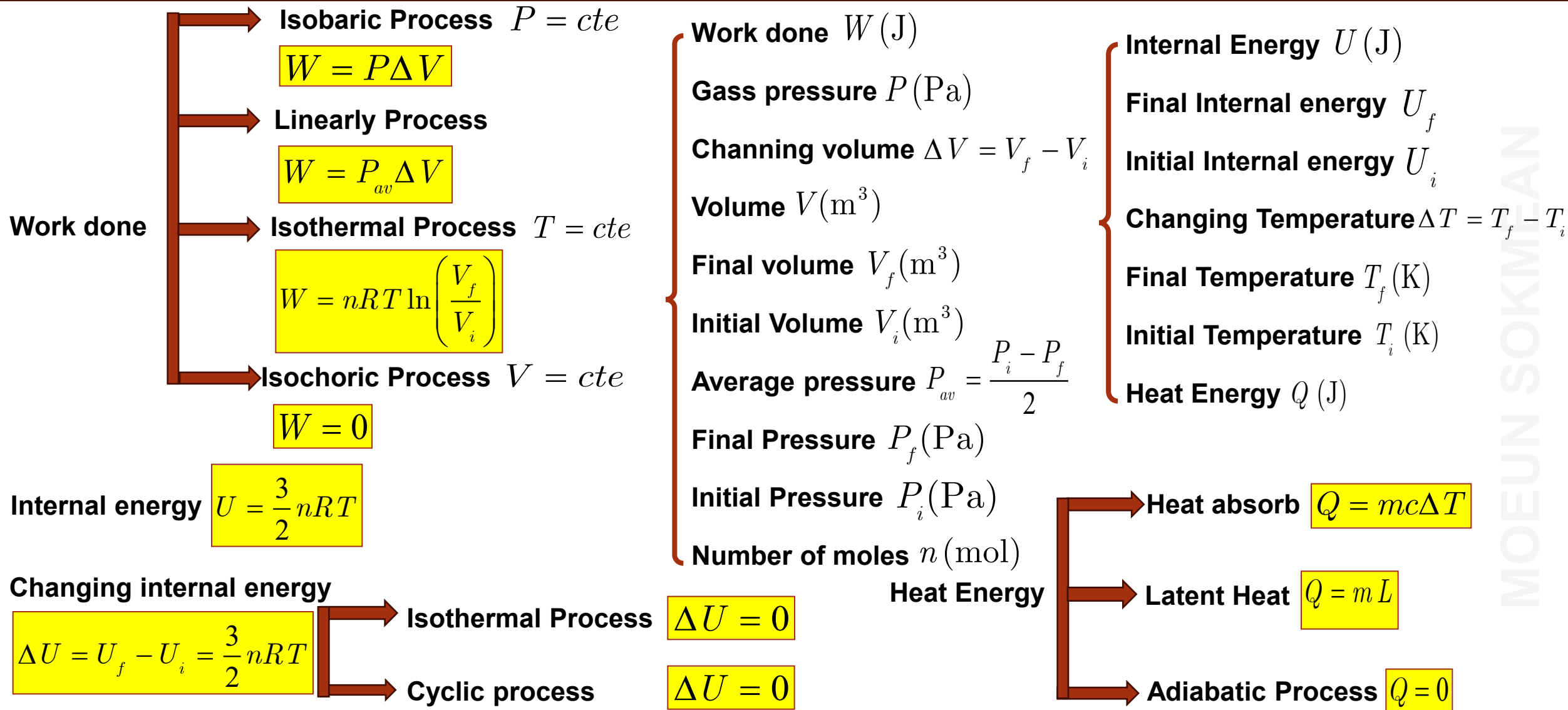


$$T_K = T_C + 273$$



Lesson1

Kinetic Theory of Ideal Gases



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Lesson2 First Law of Thermodynamic

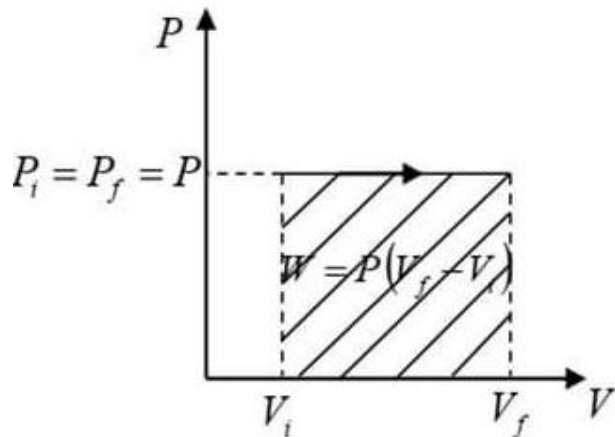


Work done and (P-V) Diagram

Isobaric Process

$$W = P\Delta V = P(V_f - V_i)$$

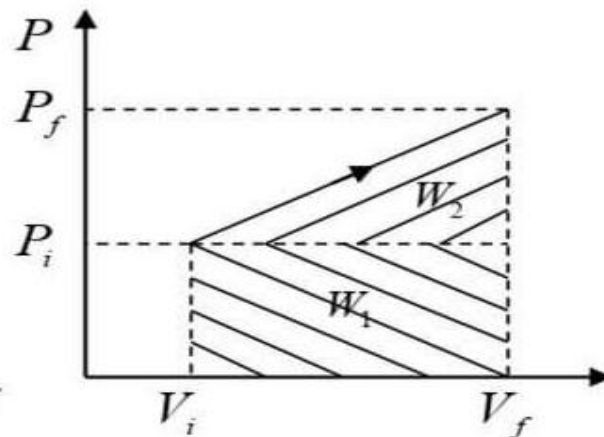
Diagram (P-V)



Linearly Process

$$W = P\Delta V = P(V_f - V_i)$$

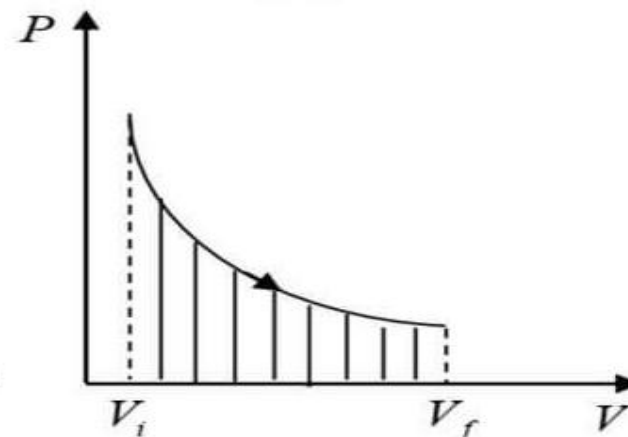
Diagram (P-V)



Isothermal Process

$$W = nRT \ln \left(\frac{V_f}{V_i} \right)$$

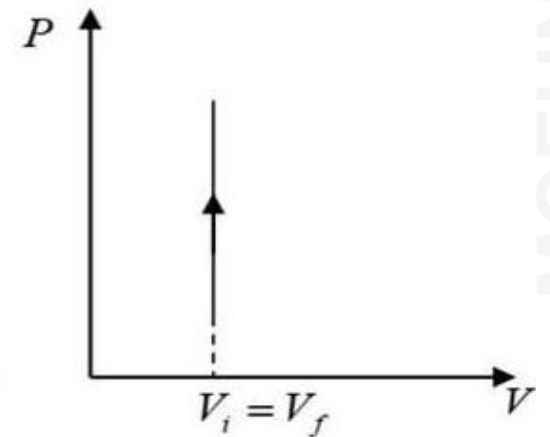
Diagram (P-V)



Isochoric Process

$$W = 0$$

Diagram (P-V)



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Lesson2 First Law of Thermodynamic



First's law of thermodynamics: $Q = W + \Delta U$

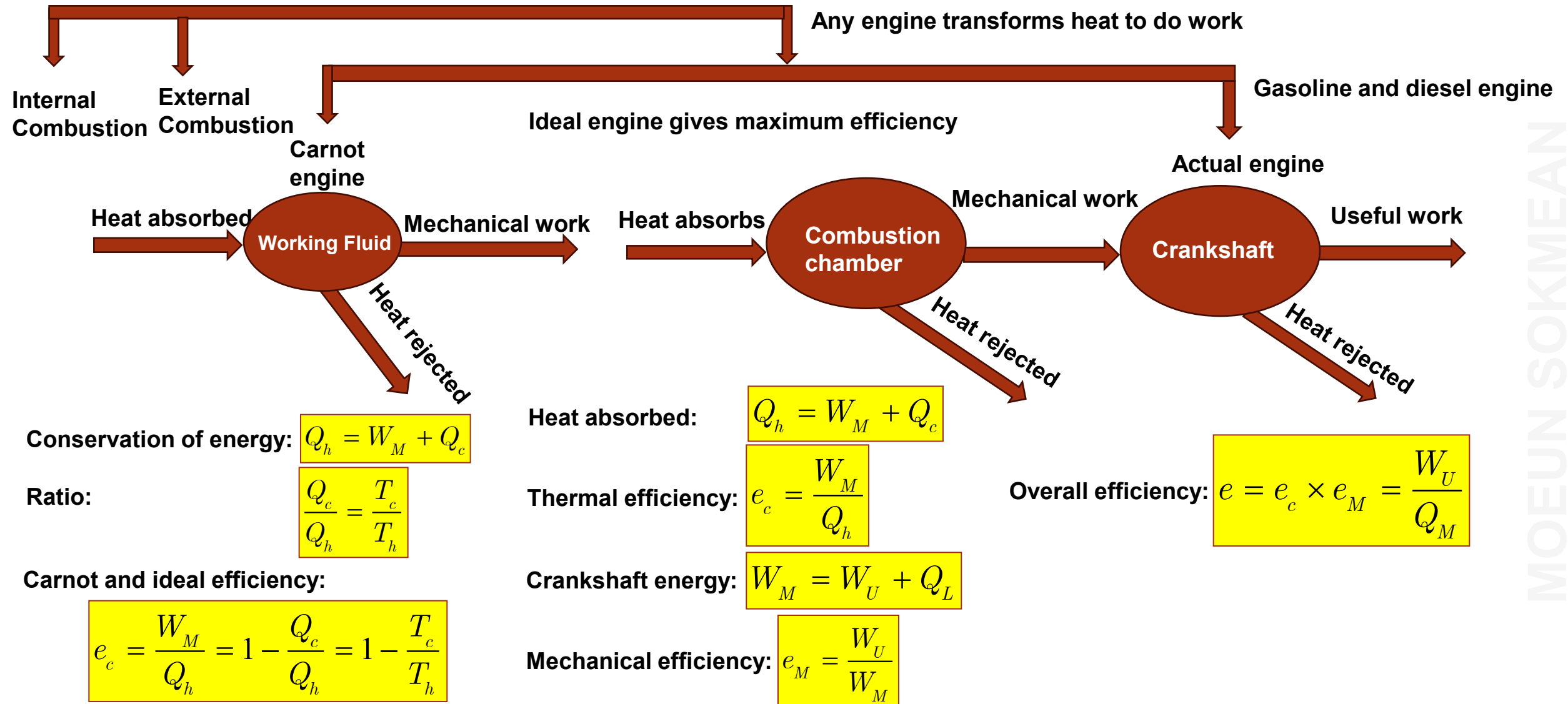
A system is defined as a set of objects chosen for study, considered in relation to its surroundings.

Sign Convention in first law thermodynamic $Q, W, \Delta U$

Heat energy Q		Work done W		Changing in internal energy ΔU	
System	Sign	System	Sign	System	Sign
Absorbed	+	Does work	+	Increased in internal energy	+
Expelled	–	Received work	–	Decreased in internal energy	–
Adiabatic Process	0	Isochoric Process	0	Isothermal Process and Cyclic Process	0



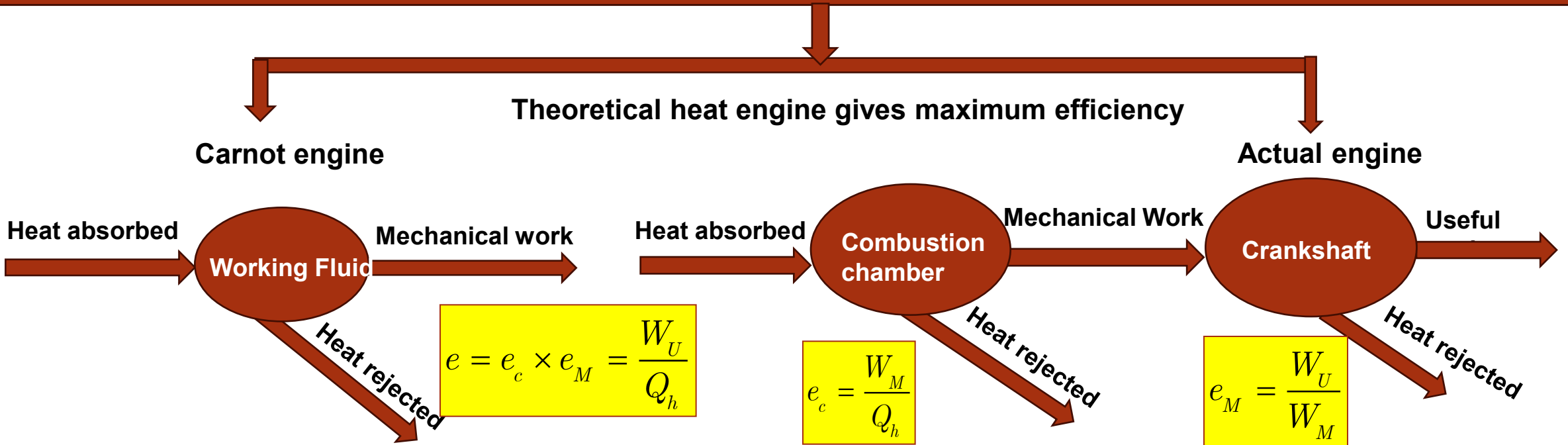
Lesson3 Heat Engine



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Lesson3 Heat Engine



Heat absorbed by working fluid Q_h (J)

Heat rejected by work fluid Q_c (J)

Mechanical work W_M (J)

Carnot or ideal efficiency

$$e_c = \frac{W_M}{Q_h} = 1 - \frac{Q_c}{Q_h} = 1 - \frac{T_c}{T_h}$$

Heat absorbed by a combustion chamber is Q_h (J)

Heat rejected by a combustion chamber is Q_c (J)

Mechanical or energy delivered by a crankshaft is W_M (J)

Mechanical work is W_M (J)

Heat rejected by a crankshaft is Q_L (J)



Superposition and Standing Wave



Sinusoidal wave is the wave that has constant amplitude and period.

Equation

$$y = a \sin(\omega t + \phi)$$

Wave equation

$$\begin{aligned} y &= a \sin(\omega t - kx) \\ y &= a \sin(kx - \omega t) \\ y &= a \sin(\omega t + kx) \\ y &= a \sin(kx + \omega t) \end{aligned}$$

Right direction

Left direction

Angular frequency:

$$\omega = \frac{2\pi}{T}, k = \frac{2\pi}{\lambda}$$

Wave speed:

$$v = \frac{x}{t} = \frac{\lambda}{T} = \frac{\omega}{k}$$

Period and frequency:

$$T = \frac{1}{f}, f = \frac{1}{T}$$

Standing wave is a wave formed by two or more waves propagate in a medium.

Method1: $(a_1 = a_2 = a)$

$$\begin{aligned} y_1 &= a \sin(\omega t + \phi_1); y_2 = a \sin(\omega t + \phi_2) \\ y &= y_1 + y_2 = 2a \cos\left(\frac{\phi_1 - \phi_2}{2}\right) \sin\left(\omega t + \frac{\phi_1 + \phi_2}{2}\right) \end{aligned}$$

Method2: $y = y_1 + y_2 = a \sin(\omega t + \phi)$

- $a = \sqrt{a_x^2 + a_y^2}$
- $a_x = a_1 \cos \phi_1 + a_2 \cos \phi_2 + \dots + a_n \cos \phi_n$
- $a_y = a_1 \sin \phi_1 + a_2 \sin \phi_2 + \dots + a_n \sin \phi_n$
- $a_x > a_y > 0 \Rightarrow \phi = \theta \left(\theta = \tan^{-1} \left| \frac{a_y}{a_x} \right| \right)$
- $a_x > 0, a_y < 0 \Rightarrow \phi = -\theta$
- $a_x < 0, a_y > 0 \Rightarrow \phi = \pi - \theta$
- $a_x < 0, a_y < 0 \Rightarrow \phi = \pi + \theta$

Standing wave is a wave formed by two or more waves propagate in a medium in the opposite direction.

Standing wave equation:

$$y = 2a \sin(kx) \cos(\omega t)$$

Amplitude: $A = 2a \sin(kx)$

Minimum amplitude

$$\begin{aligned} A &= 0 \Leftrightarrow \sin(kx) = 0 \\ \Rightarrow x &= n \frac{\lambda}{2} (n = 0, 1, 2, \dots) \end{aligned}$$

Maximum Amplitude

$$\begin{aligned} A &= \pm 2a \Leftrightarrow \sin(kx) = \pm 1 \\ \Rightarrow x &= (2n + 1) \frac{\lambda}{2} (n = 0, 1, 2, \dots) \end{aligned}$$



Lesson1 Magnetic Field



Magnetic field in current carrying wire

Magnetic force or Lorentz force

Stright wire

Circular wire

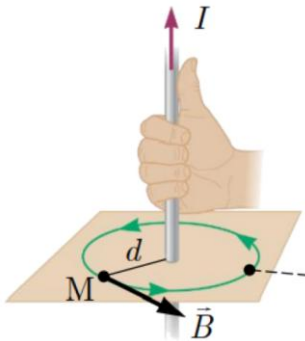
On a charge in the magnetic field

On current wires

Over a distance d from the wire

$$B = \frac{\mu_0 I}{2\pi d}$$

Direction

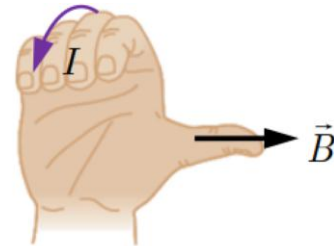


Magnetic field lines

Coils

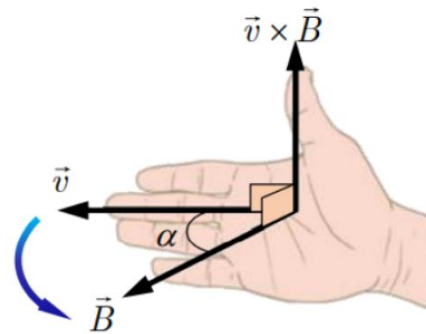
$$B = \frac{\mu_0 NI}{2R}$$

Direction



Solenoid

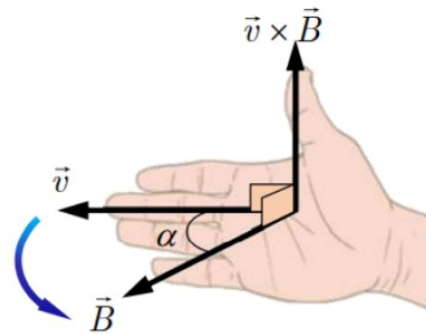
$$B = \frac{\mu_0 NI}{l}$$



Lorentz force

$$F = |q| v B \sin(\vec{v}, \vec{B})$$

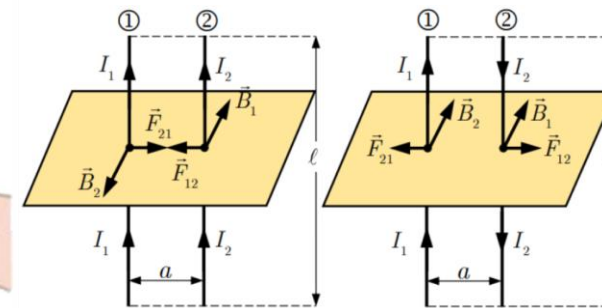
Direction



Two parallel wires

$$F = \frac{\mu_0 I_1 I_2 l}{2\pi a}$$

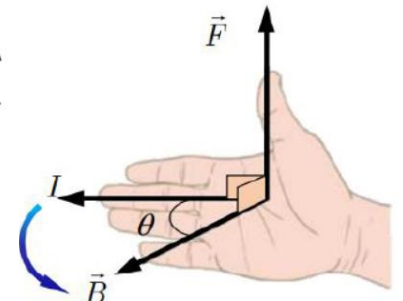
Direction



Wire in a uniform magnetic field

$$F = IB\ell \sin(\vec{\ell}, \vec{B})$$

Direction



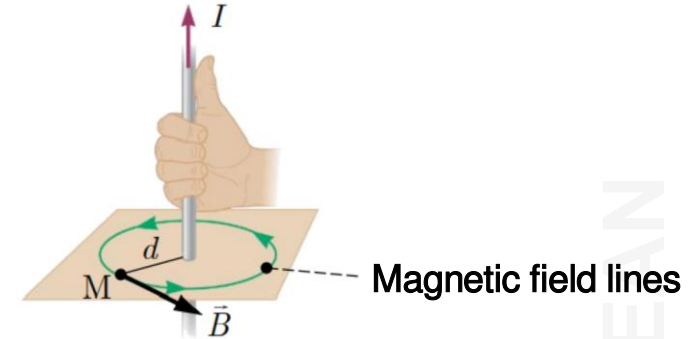
Lesson1 Magnetic Field



❖ Magnetic field due to a straight current carrying wire

- Point of application at a point located at a perpendicular distance d from the wire.
- Direction is perpendicular to the plane containing the wire.
- Direction: Thumb points in current, fingers curl in field direction.

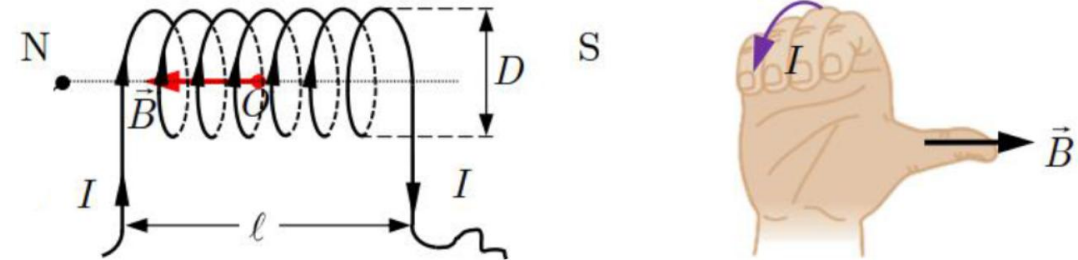
○ Magnitude: $B = \frac{\mu_0 I}{2\pi d}$



❖ Magnetic field due to a circular wire

- Point of application is taken at the center of the circular loop.
- Direction is directed perpendicular to the plane of the loop.
- Direction: Fingers follow current, thumb points along B.

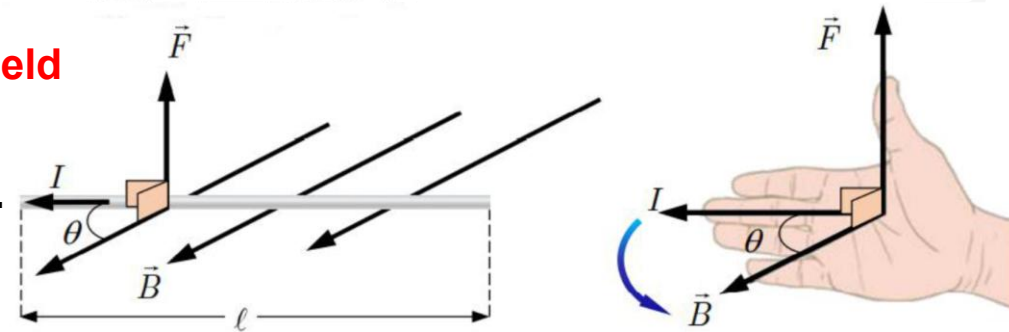
○ Magnitude $B = \frac{\mu_0 I}{2\pi R}$ (Coil) $B = \frac{\mu_0 NI}{l}$ (Solenoid)



❖ Magnetic force on a current carrying wire in a uniform magnetic field

- Point of application at the midpoint of the wire.
- Direction is perpendicular to plane of current I and magnetic field B .
- Direction: Palm along I , curl to B , thumb shows F .

○ Magnitude $F = IBl \sin \theta, \theta = (\vec{I}, \vec{B})$



Lesson1 Magnetic Field

❖ Magnetic field formed by two parallel wire

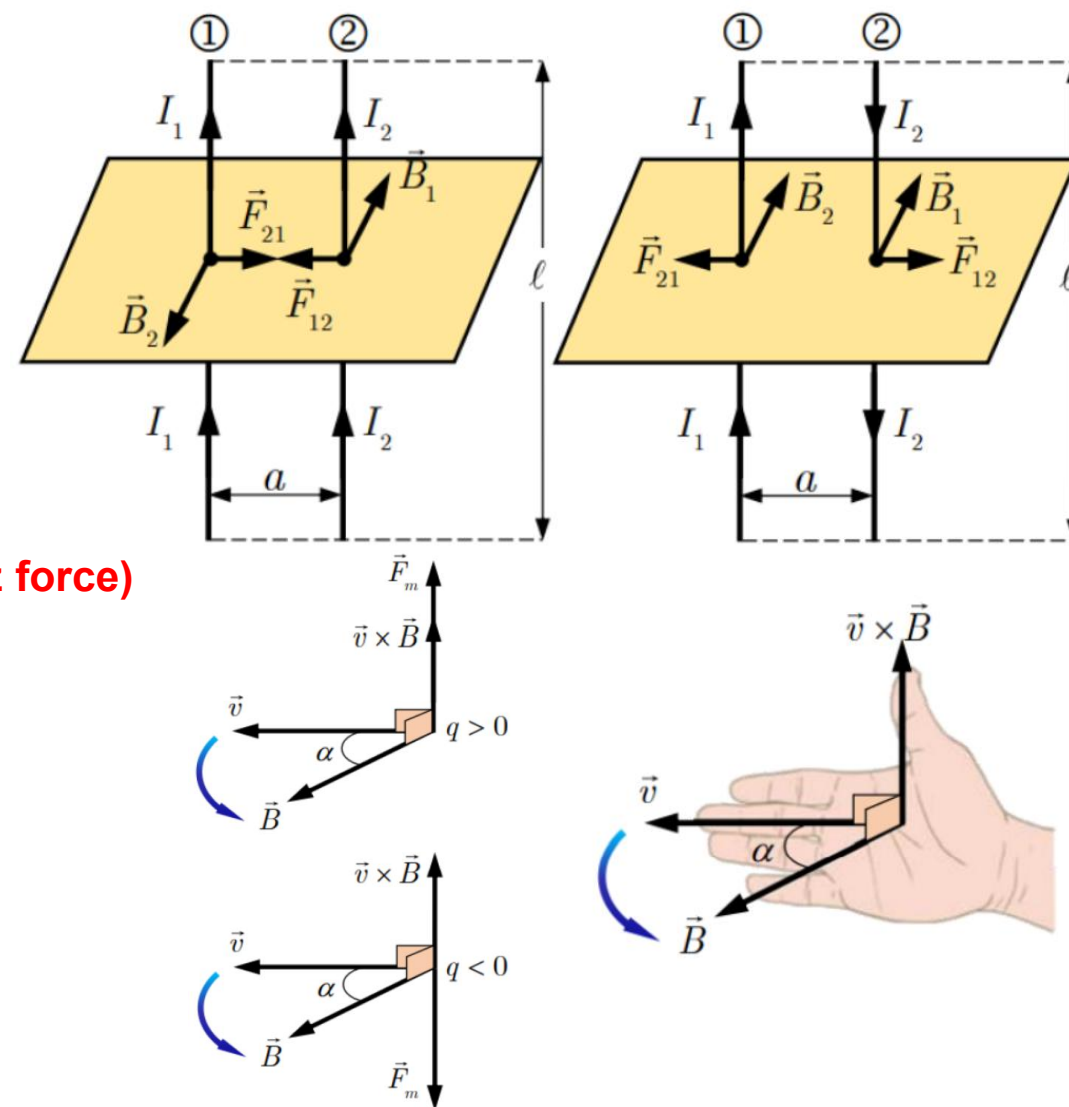
- Point application at the center of each wire
- Direction is perpendicular to I - B plane..
- Direction: Palm along I , fingers curl B , thumb shows F .
- ✗ If $I_1 \uparrow \uparrow I_2$, the force between them is attractive.
- ✗ If $I_1 \uparrow \downarrow I_2$, the force between them is repulsive.

○ Magnitude $F_{12} = F_{21} = F = \mu_0 \frac{I_1 I_2 \ell}{2\pi a}$

❖ Magnetic field experiencing on a charged particle (Lorentz force)

- Point of application at the location of the charged particle.
- Direction is Perpendicular to velocity- B plane.
- Direction: Fingers along v , curl to B , thumb shows F
- ✗ If $q > 0 \Rightarrow \vec{F} \uparrow \uparrow (\vec{v} \times \vec{B})$
- ✗ If $q < 0 \Rightarrow \vec{F} \uparrow \downarrow (\vec{v} \times \vec{B})$

○ Magnitude $F = |q|vB \sin \theta, \theta = (\vec{v}, \vec{B})$



Lesson1 Magnetic Field



■ SI Unit of each variable

Magnetic Field Produced by an Electric Current

■ Note:

$$\mu_r = \frac{\mu}{\mu_0} \text{ Relative permeability}$$

$$\mu_0 = 4\pi \times 10^{-7} \text{ Tm/A Permeability of free space}$$

$$\mu = \mu_0 \times \mu_r \text{ Permeability of the material}$$

Electromagnetic force

Straight current wire:

Over distance d from wire

$$B = \mu_0 \frac{I}{2\pi d}$$

Circular wire

Coils:

$$B = \mu_0 \frac{NI}{2R}$$

Solenoid:

$$B = \mu_0 \frac{NI}{\ell}$$

Magnetic field B (T)

Permeability of free space

Current I (A) $\mu_0 = 4\pi \times 10^{-7} \text{ Tm/A}$

Distance from wire d (m)

Number of turns N

Radius of a loop R (m)

Length of the solenoid ℓ (m)

A wire in magnetic field

$$F = IBl \sin(\vec{I}, \vec{B})$$

Two parallel wires

$$F = \mu_0 \frac{I_1 I_2 l}{2\pi a}$$

On current

On a charged in magnetic field

$$F = |q| v B \sin(\vec{v}, \vec{B})$$

Magnetic force F (N)

Magnetic field B (T)

Current I (A)

Length of wire l (m)

Distance between two wires a (m) v (m/s)

Electric charges q (C) speed of a charge

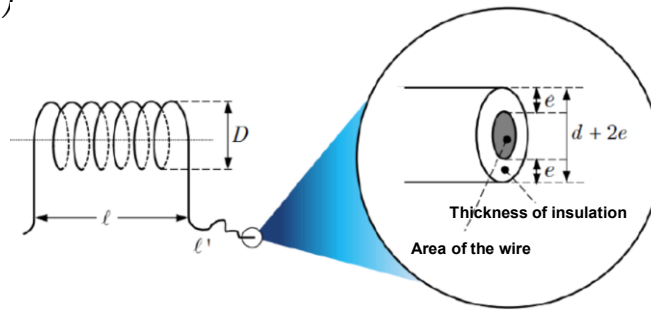
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❖ How to determine the number of turns and resistance of the solenoid



- Number of turns N
- Length of the solenoid l (m)
- Diameter of the solenoid D (m)
- Radius of a solenoid R_s (m)
- Area of a solenoid A (m²)
- Number of layers x
- Resistance of the wire ρ (Ωm)
- Resistivity of the wire l' (m)
- Diameter of the wire d (m)
- Radius of wire r (m)
- Area of the wire A' (m²)
- Thickness of the insulation e (m)



- Number of turns

$$N = \frac{l'}{\pi D}$$

- Number of turns

$$N = \frac{l}{d + 2e} x$$

- In case

$$(x = 1; e \neq 0) N = \frac{l}{d + 2e} x$$

- In case

$$(x > 1; e = 0) N = \frac{l}{d} x$$

- In case

$$(x = 1; e = 0) N = \frac{l}{d}$$

- Resistance of the wire (solenoid)

$$R = \rho \frac{l'}{A'}$$

- Length of the wire

$$l' = \pi D N (m)$$

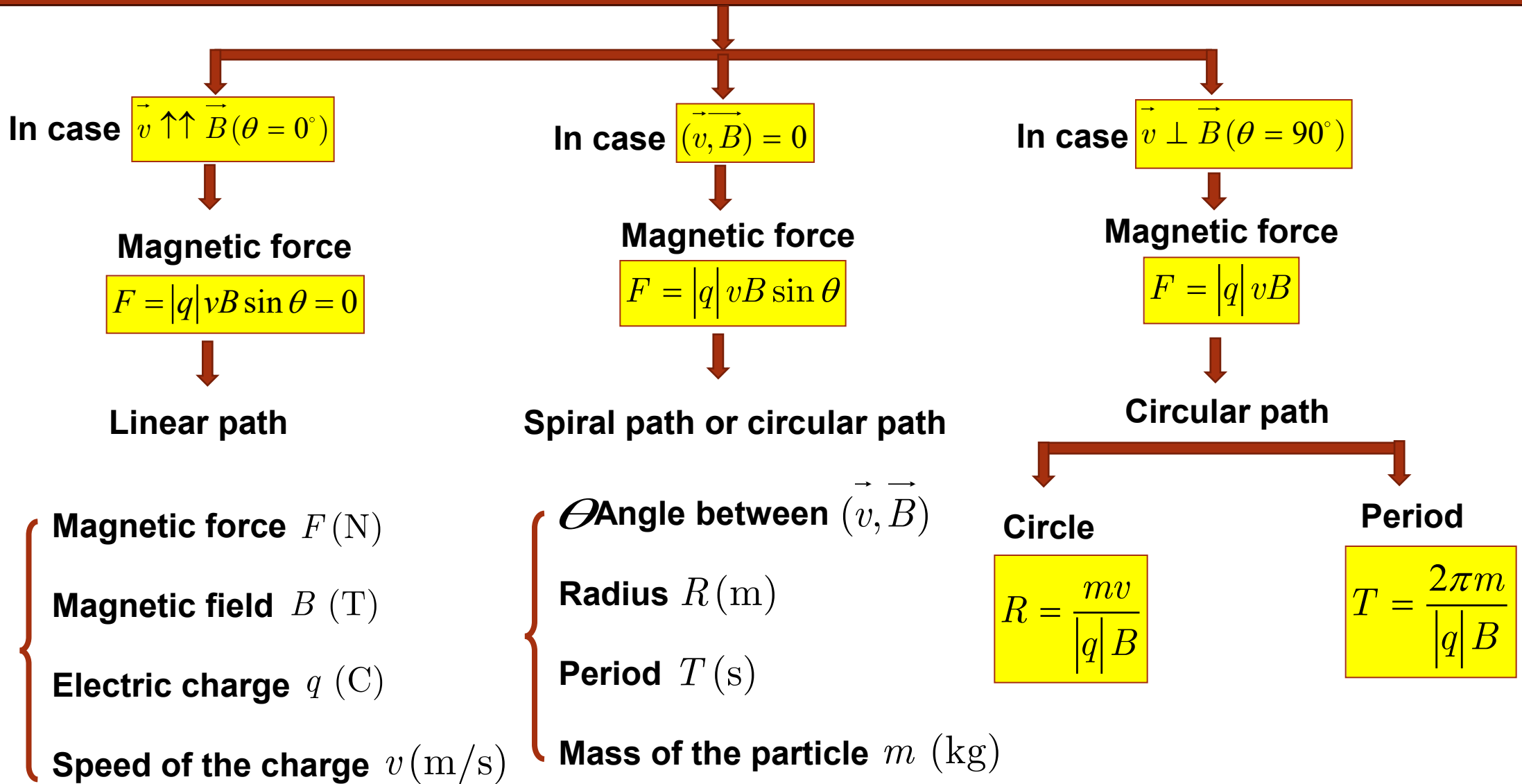
- Area of the wire

$$A' = \frac{\pi d^2}{4} (m^2)$$

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❖ The path of the electric charged moved in the uniform magnetic field



Lesson2: Induced emf

→ ❖ Experiment



Magnetic flux



Change in magnetic flux

$$\Phi = NBA \cos(\vec{B}, \vec{n})$$

$$\Delta\Phi = \Phi_f - \Phi_i$$



Induce emf

Faraday's law

A moving conductor in a magnetic field

$$E = vBl \sin(\vec{v}, \vec{B})$$

Generator

$$e = NBA\omega \sin(\omega t)$$

Average

$$E = -\frac{\Delta\Phi}{\Delta t}$$
$$|E| = \left| -\frac{\Delta\Phi}{\Delta t} \right|$$

Magnitude

$$e = -\frac{d\Phi}{dt}$$
$$|e| = \left| -\frac{d\Phi}{dt} \right|$$

Closed circuit



❖ Induce current

Direction



Lenz's law

Magnitude



Ohm's law

$$I = \frac{|E|}{R}$$

Reversibility



Electric motors



❖ SI Unit of each variable in the formula



Magnetic flux: $\Phi = NBA \cos(\vec{B}, \vec{n})$

Magnetic flux Φ (Wb)

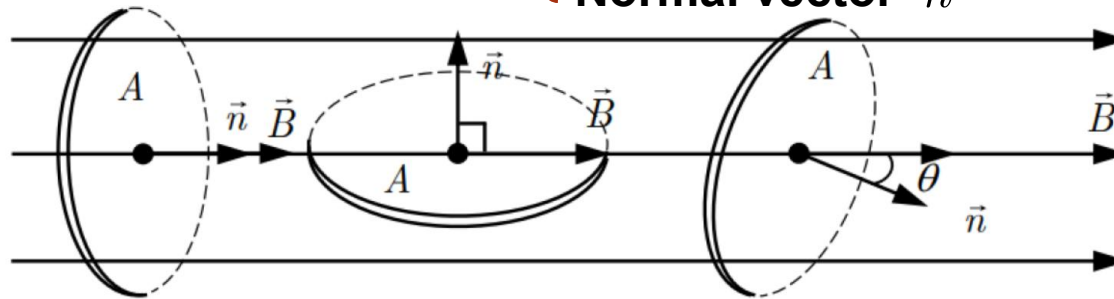
Magnetic field B (T)

Number of turns N

Area of a closed circuit A (m²)

Angle between $\theta = (\vec{B}, \vec{n})$

Normal vector $\vec{n}$



$$\Phi_m = BA \text{ because } \theta = 0^\circ \quad \Phi_m = BA \cos \theta \quad \Phi_m = 0 \text{ because } \theta = 90^\circ$$

Changing in magnetic flux: $\Delta\Phi = \Phi_f - \Phi_i$

Changing in magnetic flux $\Delta\Phi$

Final magnetic flux Φ_f

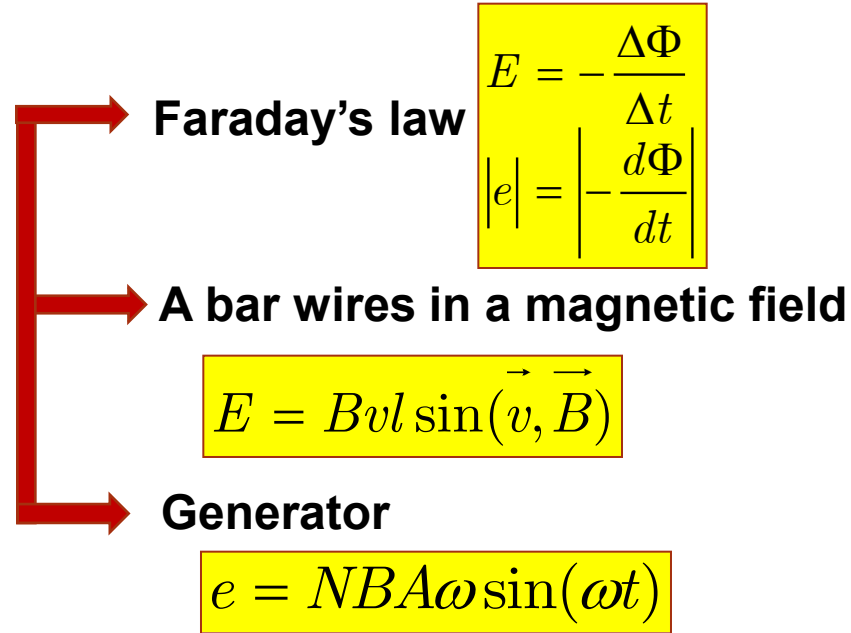
Initial magnetic flux Φ_i



❖ SI Unit of each variable in the formula



Induce electromotive force



Electromotive force E (V)

Changing in magnetic flux $\Delta\Phi = \Phi_f - \Phi_i$

Duration $\Delta t = t_f - t_i$

Derivative of flux $\frac{d\Phi}{dt}$

Induced current

$$I = \frac{|E|}{R}$$

Induced emf E (V)

Induced current I (A)

Resistance R (Ω)

Induced emf e (V)

Magnetic field B (T)

Number of turns N

Electromotive force E (V)

Magnetic field B (T)

Speed of the wire v (m/s)

Length of the wire l (m)

Angle between $\theta = (\vec{v}, \vec{B})$

Area A (m^2)

Angular frequency ω (m/s)

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Lesson3 Self Induction



Changing current in a coil

$$\Delta I, dI$$

Relationship between current and magnetic flux

$$\Phi = LI$$

Changing magnetic flux in a coil

$$\Delta\Phi = L\Delta I$$

Self induced emf

Magnitude

$$E = -L \frac{\Delta I}{\Delta t}$$

Instantaneous value

$$e = -L \frac{dI}{dt}$$

Changing current $\Delta I = I_2 - I_1$

Duration $\Delta t = t_2 - t_1$

Derivative respect to time $\frac{di}{dt} (\text{A/s})$

Inductance $L (\text{H})$

Self induced electromotive force $e (\text{V})$

Changing in magnetic flux $\Delta\Phi (\text{Wb})$

Number of turns N

Area of the solenoid $A (\text{m}^2)$

Length of the solenoid $l (\text{m})$

Permeability of free space $\mu_0 = 4\pi \times 10^{-7} \text{Tm/A}$

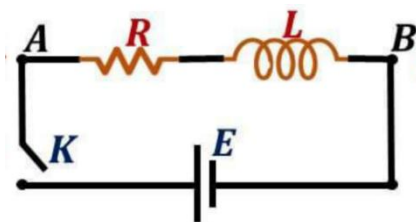
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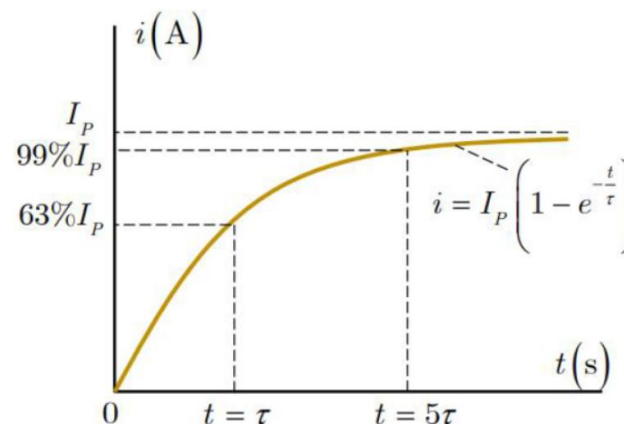
Lesson3 Self Induction



RL Circcuit



When the switch is closed



Voltage Equation

$$V_{AB} = Ri + L \frac{di}{dt}$$

Differential equation of current

$$I_P = i + \tau \frac{di}{dt}$$

Studies at instance t

Instantaneous current

$$i = I_P \left(1 - e^{-\frac{t}{\tau}} \right)$$

Time constant $\tau = \frac{L}{R}$

$$t = \tau$$

$$t = 5\tau$$

$$\tau = \infty$$

$$i = 65\% I_P$$

$$i = 99\% I_P$$

$$i = i_P$$

Peak current

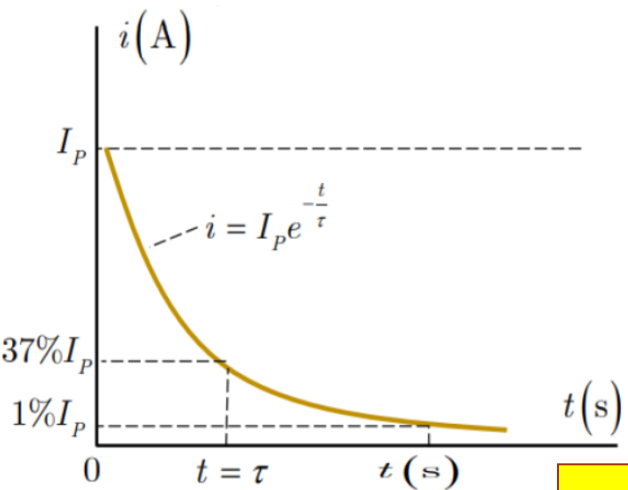
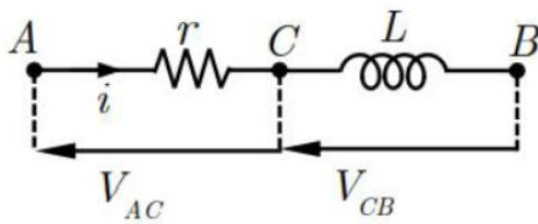
$$I_P = \frac{E}{R}$$



Lesson3 Self Induction



RL Circuit



When the switch is opened

Voltage equation $0 = Ri + L \frac{di}{dt}$

Differential equation of current $0 = i + \tau \frac{di}{dt}$

Studied at instance t

Instantaneous current

$$i = I_P e^{-\frac{t}{\tau}}$$

Time constant:

$$\tau = \frac{L}{R}$$

$$t = 0$$

$$t = \tau$$

$$\tau = \infty$$

$$i = I_P$$

$$i = 36\% I_P$$

$$i = 0$$

Peak current

$$I_P = \frac{E}{R}$$



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Lesson3 Self Induction



RL Circuit



When the switch is closed

Conservation of energy in LC the oscillation



$$E_{CL} = E_C + E_L = \frac{1}{2}CV^2 + \frac{1}{2}Li^2 = \text{constant}$$

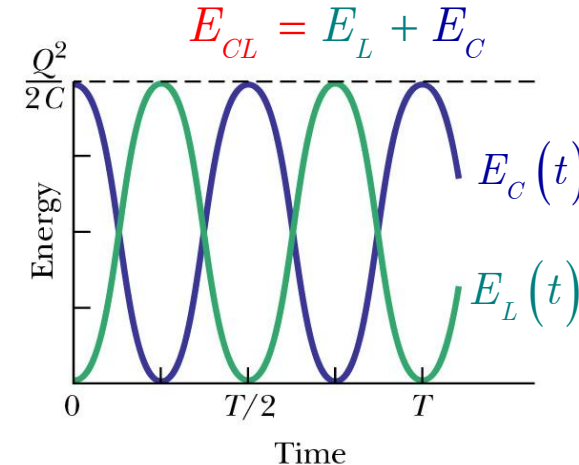
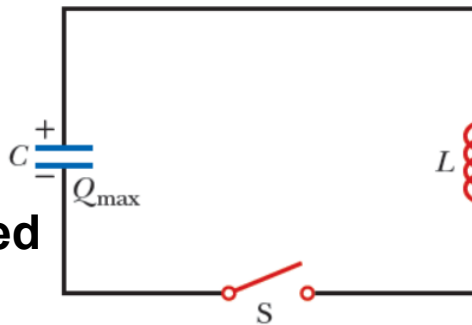


**Energy stored
in a capacitor**

$$E_C = \frac{1}{2}CV^2$$

**Energy stored
in an inductor**

$$E_L = \frac{1}{2}Li^2$$



$$E_{CL} = E_{Cm} = \frac{1}{2}CV_m^2$$

$$i = 0; V = V_m$$

$$i = i_m; V = 0$$

Maximum energy stored in a capacitor

Maximum energy stored in an inductor

$$E_{CL} = E_{Lm} = \frac{1}{2}LI_m^2$$

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RL Circuit

Closed the switch

Voltage equation

$$V_C'' + \omega^2 V_C = 0$$

Angular frequency

$$\omega_0 = \frac{1}{\sqrt{CL}}$$

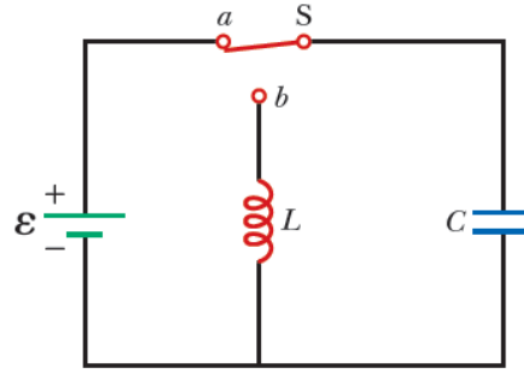
Period

$$T = 2\pi\sqrt{CL}$$

$$f_0 = \frac{1}{T_0}$$

Angular frequency

$$f = \frac{1}{2\pi\sqrt{CL}}$$



Instantaneous voltage $V_C = V_m \cos(\omega_0 t + \varphi_0)$

$$q_C = CV_C \rightarrow V_m = E$$

Instantaneous charge $q_c = q_m \cos(\omega_0 t + \varphi_0)$

$$i_c = \frac{dq_C}{dt}$$

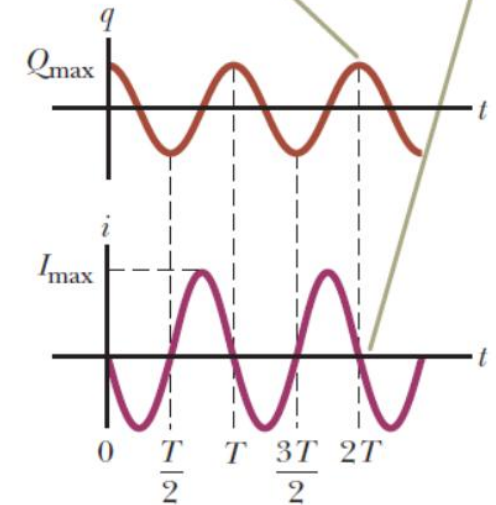
Instantaneous current $i_c = i_m \sin(\omega_0 t + \varphi_0)$

Alternatively $i_c = i_m \cos(\omega_0 t + \varphi_0 + \frac{\pi}{2})$

$$q_m = CV_m$$

$$I_m = \omega_0 q_m$$

The charge q and the current i are 90° out of phase with each other.



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Lesson4 Alternating Current



1. General State

1.1. Definition of direct current

○ Alternating current is the current that changes its direction two times in one cycle.

○ The simplest current is the sinusoidal current. Sinusoidal current is in the form: $i = i_m \sin(\omega t + \varphi_0)$

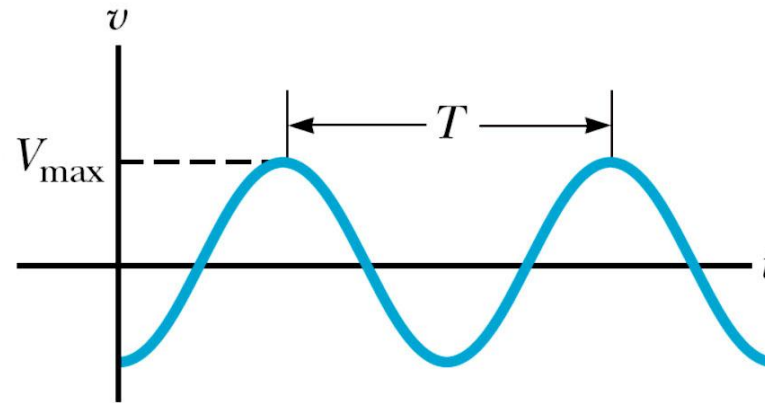
✎ $i(A)$ Instantaneous current at instant t

✎ $i_m(A)$ Maximum current or amplitude

✎ $\omega(rad/s)$ Angular frequency

✎ $(\omega t + \varphi)$ Instant phase

✎ $\omega_0(rad/s)$ Initial phase at $t = 0$



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Lesson4 Alternating Current



1.2. Value table for current and voltage in the circuit

- In an alternating current (AC) circuit, the values of current and voltage can be expressed in three different forms.

	Instantaneous value	Maximum value	RMS Values
Current	i	I_m	$I = \frac{I_m}{\sqrt{2}}$
Voltage	v	V_m	$V = \frac{V_m}{\sqrt{2}}$

- RMS current or voltage produces same heating as DC in same duration.
- Ohm's law applied in alternating current: $V = ZI$, where Z is the impedance."

$$I = \frac{V}{Z} \quad I_m = \frac{V_m}{Z}$$



Lesson4 Alternating Current



1. Resistance in R, L, C Alternating Circuits

1.1 For Resistor

○ We have: $v = V_m \sin(\omega t)$ (1)

○ Ohm's law $I_m = \frac{V_m}{Z_R}$ $Z_R = R$ where R is resistance measured in (Ω)

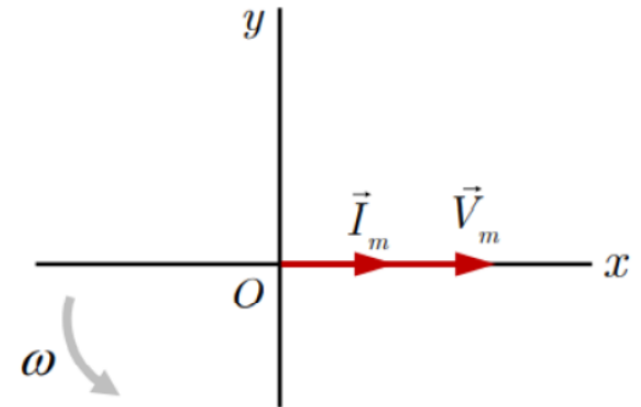
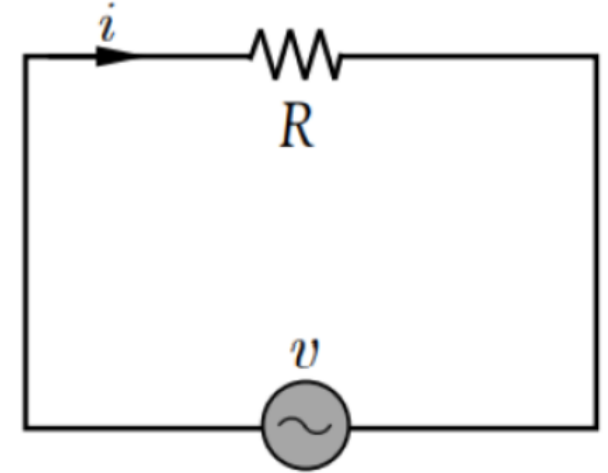
○ Instantaneous current $i = i_m \sin(\omega t)$

○ From equations (1) and (2), we see that for a resistor, the current and voltage are in phase.

○ Fresnel diagram

Since $i = I_m \sin(\omega t)$ where $\phi_i = \left(\vec{I}_m, \vec{ox} \right)_{t=0} = 0$

And $v = V_m \sin(\omega t)$ where $\phi_v = \left(\vec{V}_m, \vec{ox} \right)_{t=0} = 0$



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Lesson4 Alternating Current



2.2 For capacitor

○ In alternating circuit, value of current and voltage divided into three

○ We have $v = V_m \sin(\omega t)$ (1)

○ Ohm's law $I_m = \frac{V_m}{Z_C}$ where $Z_C = \frac{1}{C\omega}$ is the impedance of the capacitor (Ω).

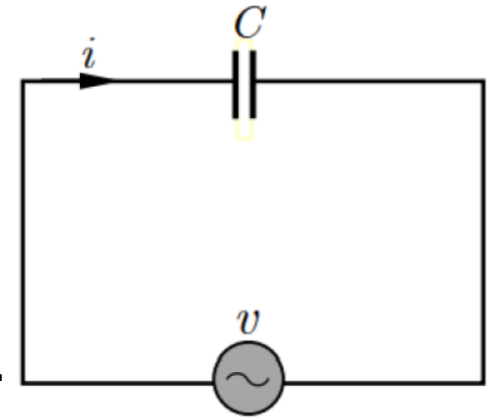
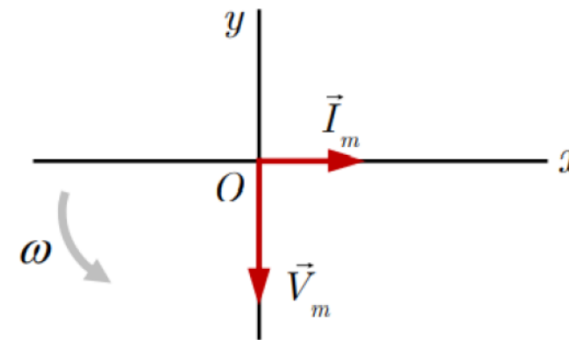
○ Instantaneous current: $i = I_m \cos(\omega t) = I_m \sin\left(\omega t + \frac{\pi}{2}\right)$ (2)

○ From equations (1) and (2), we see that the current leads the voltage by 90° . $\frac{\pi}{2} \left(\varphi_i = \varphi_v - \frac{\pi}{2} \right)$

○ Fresnel diagram

Since $i = I_m \sin\left(\omega t + \frac{\pi}{2}\right)$ Where $\varphi_i = \left(\vec{I}_m, \vec{ox}\right)_{t=0} = \frac{\pi}{2}$

And $v = V_m \sin(\omega t)$ Where $\varphi_v = \left(\vec{I}_m, \vec{ox}\right)_{t=0} = 0$



Lesson4 Alternating Current



2.3 For inductor

○ We have: $i = I_m \sin(\omega t)$ (1)

○ Ohm's law $I_m = \frac{V_m}{Z_L}, Z_L = L\omega$ where, Z_L is the impedance (Ω).

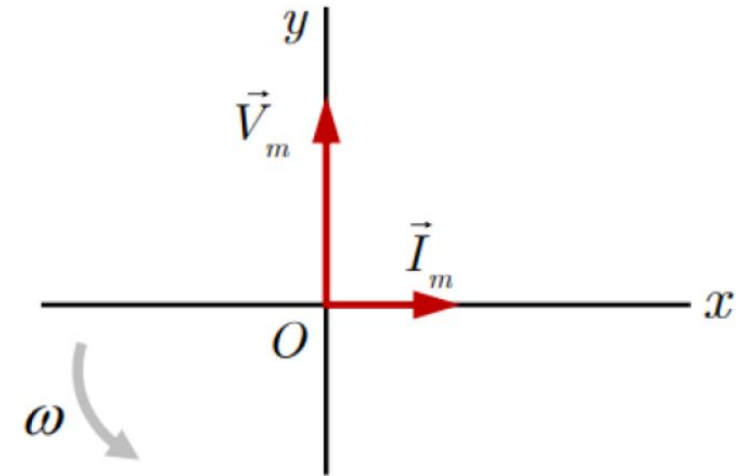
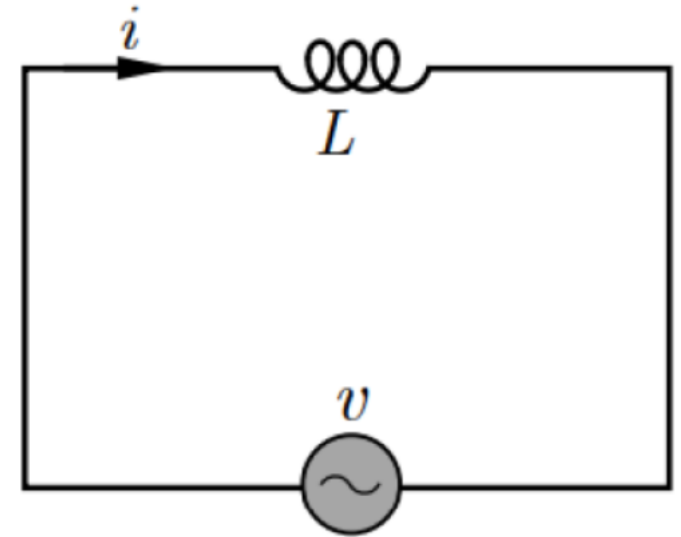
○ Instantaneous current: $v = V_m \cos(\omega t) = V_m \sin\left(\omega t + \frac{\pi}{2}\right)$ (2)

○ From equations (1) and (2), we see that the current lags the voltage by 90° .

○ Fresnel

Since $i = i_m \sin\left(\omega t - \frac{\pi}{2}\right)$ Where, $\varphi_i = \left(\vec{I}_m, \vec{ox}\right)_{t=0} = -\frac{\pi}{2}$

And $v = V_m \sin(\omega t)$ Where, $\varphi_v = \left(\vec{V}_m, \vec{ox}\right)_{t=0} = 0$



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Lesson4 Alternating Current

3. RLC Series Circuit in Alternating current

3.1. In case of RLC series circuit:

- We have instantaneous current: $i = I_m \sin(\omega t)$ (1)
- Construct Fresnel diagram $\vec{V}_{Rm}, \vec{V}_{Lm}, \vec{V}_{Cm}$ relative to $\vec{I}_m$
- From the Fresnel diagram on the right-hand side

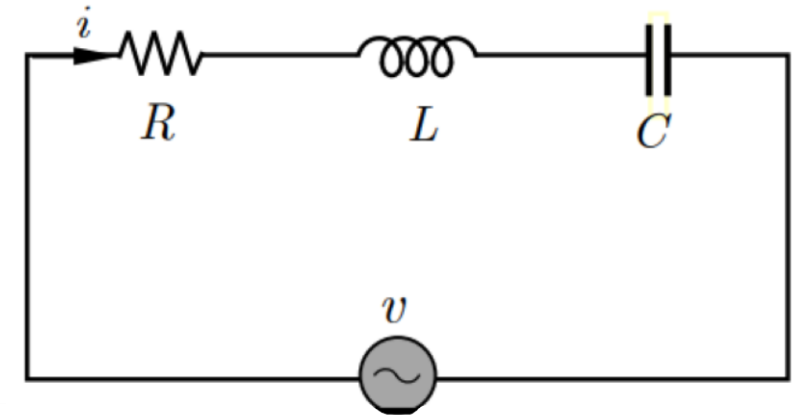
❖ Maximum voltage V_m and rms voltage V in RLC series circuit

$$\vec{V}_m = \vec{V}_{Rm} + \vec{V}_{Cm} + \vec{V}_{Lm} \Leftrightarrow V_m^2 = V_{Rm}^2 + |V_{Lm}^2 - V_{Cm}^2|$$

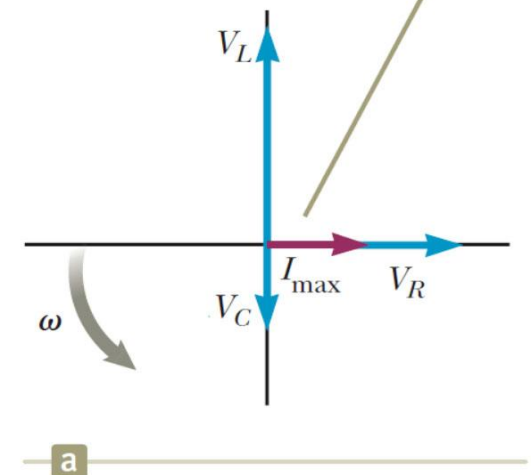
$$V_m = \sqrt{V_{Rm}^2 + |V_{Lm}^2 - V_{Cm}^2|} \Leftrightarrow V = \sqrt{V_R^2 + |V_L - V_C|^2}$$

❖ Impedance Z in RLC series circuit

$$V_m = ZI_m \Leftrightarrow \sqrt{(RI_m)^2 + |Z_L I_m - Z_C I_m|^2} = ZI_m \Leftrightarrow Z = \sqrt{R^2 + |Z_L - Z_C|^2}$$



The phasors of Figure 33.14 are combined on a single set of axes.



Lesson4 Alternating Current



$$Z = \sqrt{R^2 + \left(L\omega - \frac{1}{C\omega}\right)^2}$$

Phase Angle ϕ of RLC Series Circuit

$$\tan \phi = \frac{V_{Lm} - V_{Cm}}{V_{Rm}} = \frac{Z_L - Z_C}{R}$$

$$\cos \phi = \frac{V_{Rm}}{V_m} = \frac{R}{Z}$$

Phase Shift in RLC Series Circuit:

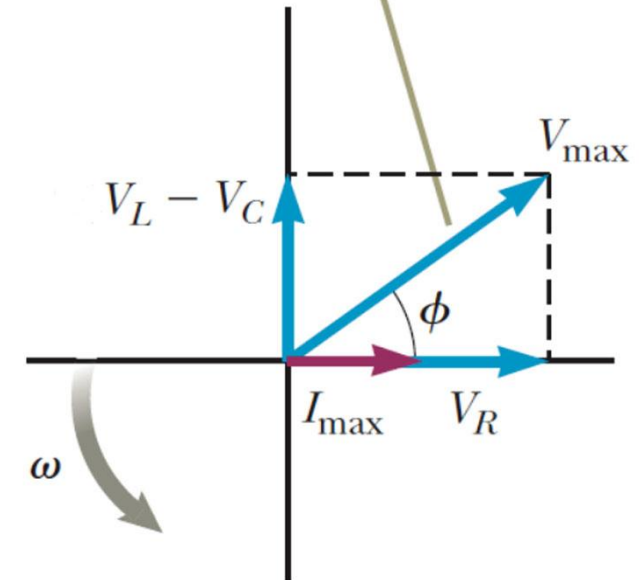
✎ If $Z_L > Z_C \Rightarrow \tan \phi > 0 \Rightarrow \phi > 0$

✎ From equation (1) and (2)

✎ Thus, the current lags the voltage, or equivalently, the voltage leads the current.

$$\begin{cases} i = I_m \sin(\omega t) & (1) \\ v = V_m \sin(\omega t + \phi) & (2) \end{cases}$$

The total voltage $V_{\max}$ makes an angle ϕ with $I_{\max}$.



b

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Lesson4 Alternating Current



✂ If $Z_L < Z_C \Rightarrow \tan \varphi < 0 \Rightarrow \varphi < 0$

From equations (1) and (2), we see that the current lags the voltage,
or equivalently, the voltage leads the current

✂ If $Z_L = Z_C \Rightarrow \tan \varphi = 0 \Rightarrow \varphi = 0$

From equations (1) and (2), we see that the current and voltage are
in phase. In this case, resonance occurs, producing maximum
current in the circuit.

$$Z = R, \varphi = 0, \cos \varphi = 1, \omega = \frac{1}{\sqrt{CL}}$$

❖ Note: They can also construct a Fresnel diagram for the RMS
voltage V and impedance Z in a series RLC circuit, since the
voltages across R, L, and C combine vectorially. Since

$$V = \frac{V_m}{\sqrt{2}}; Z = \frac{V_m}{I_m}$$



Lesson4 Alternating Current



○ Special cases RC

- $V_m = \sqrt{V_{Rm}^2 + V_{Cm}^2}$
- $Z = \sqrt{R^2 + Z_C^2}$
- $\tan \varphi = -\frac{Z_c}{R} < 0$

Hence, the current leads the voltage

○ Special case RL

- $V_m = \sqrt{V_{Rm}^2 + V_{Lm}^2}$
- $Z = \sqrt{R^2 + Z_L^2}$
- $\tan \varphi = \frac{Z_c}{R} > 0$

Hence, the current lags the voltage



Lesson4 Alternating Current



4. Average power and power factor of ac circuit

○ We have $v = V_m \sin(\omega t), i = I_m \sin(\omega t + \phi)$

○ Hence, average power $P = VI \cos(\phi)$ Where $\cos(\phi)$ is the power factor

○ **Note:**

✂ For pure resistor R, therefore

$$\phi = 0 \Rightarrow \cos \phi = 1 \Leftrightarrow P = VI = RI^2 = \frac{V^2}{R}$$

✂ For pure inductor L, therefore

$$\phi = -\frac{\pi}{2} \Rightarrow \cos \phi = 0 \Leftrightarrow P_L = 0$$

✂ For pure capacitor C, therefore

$$\phi = \frac{\pi}{2} \Rightarrow \cos \phi = 0 \Leftrightarrow P_C = 0$$

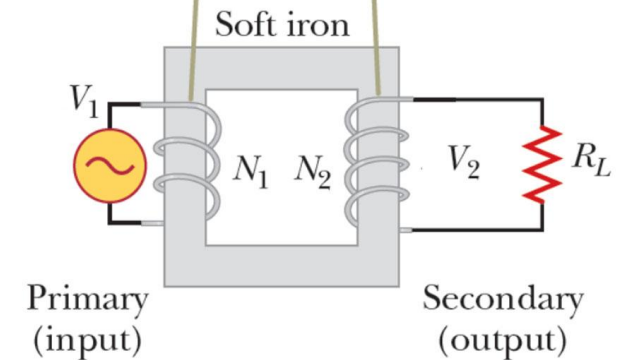
✂ In RLC series circuit, therefore

$$P = VI \cos(\phi) = ZI \times I \times \frac{R}{Z} = RI^2 (V = ZI, \cos \phi = \frac{R}{Z})$$

✂ In RLC parallel circuit, therefore

$$P = VI \cos(\phi) = V \times \frac{V}{Z} \times \frac{R}{Z} = RI^2 \left(I = \frac{V}{Z}, \cos \phi = \frac{Z}{R} \right)$$

An alternating voltage V_1 is applied to the primary coil, and the output voltage V_2 is across the resistor of resistance R_L .



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Lesson4 Alternating Current



5. Transformers

- A transformer is a device made of two parallel iron cores joined together, with wires wound around the ends to form coils.

✎ Power and equation

$$P_{e1} = P_{e2} + P_J \Leftrightarrow W_{e1} = W_{e2} + W_J$$

✎ Efficiency of a transformer:

$$Rd = \frac{P_{e2}}{P_{e1}} = \frac{W_{e2}}{W_{e1}} < 1$$

✎ Turn ratio of a transformer:

$$K = \frac{V_2}{V_1} = \frac{n_2}{n_1}$$

✎ $K > 1 \Rightarrow V_2 > V_1$ Is called step up transformer.

✎ $K < 1 \Rightarrow V_2 < V_1$ Is called step down transformer.



Q & A

Thank You!